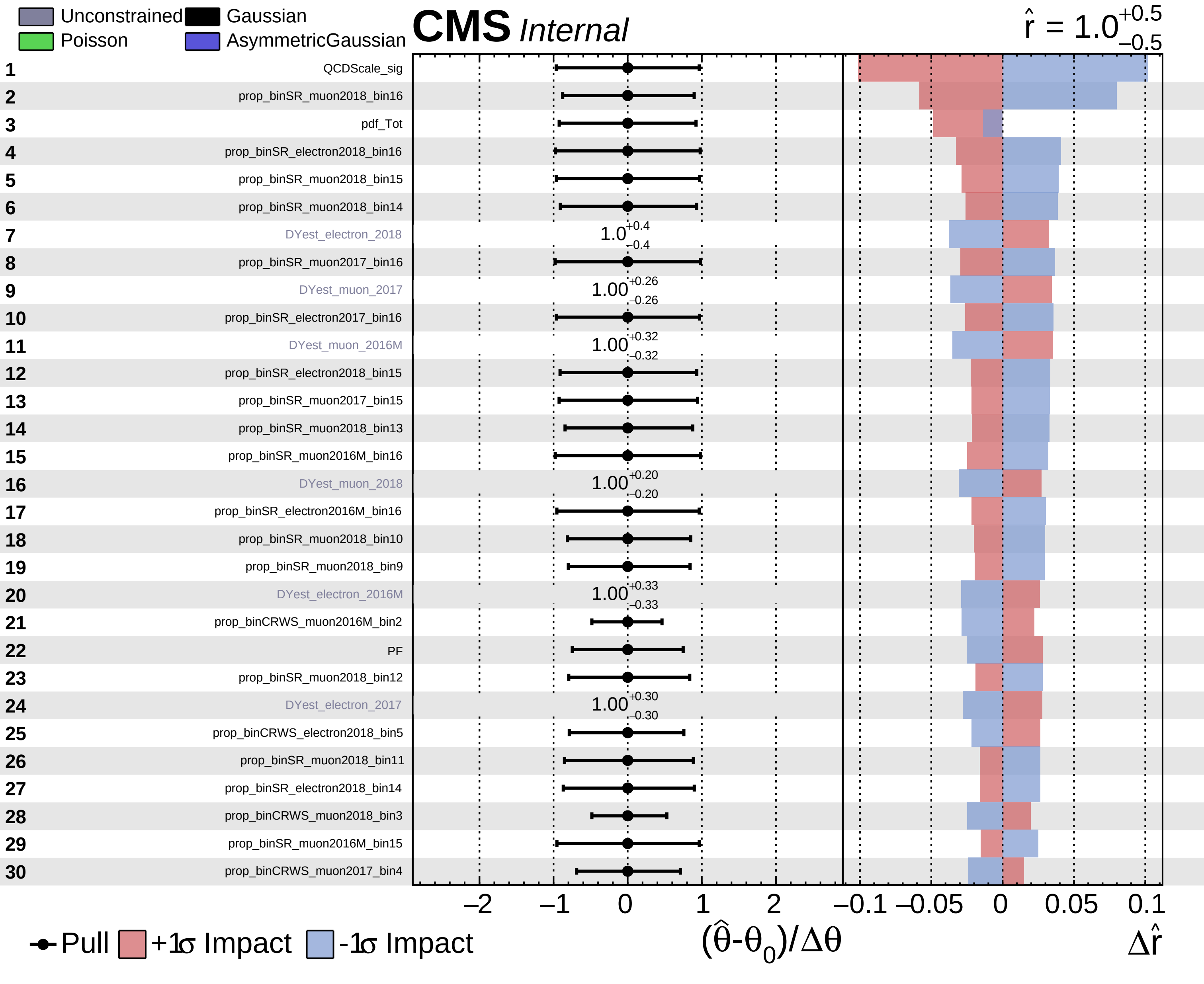
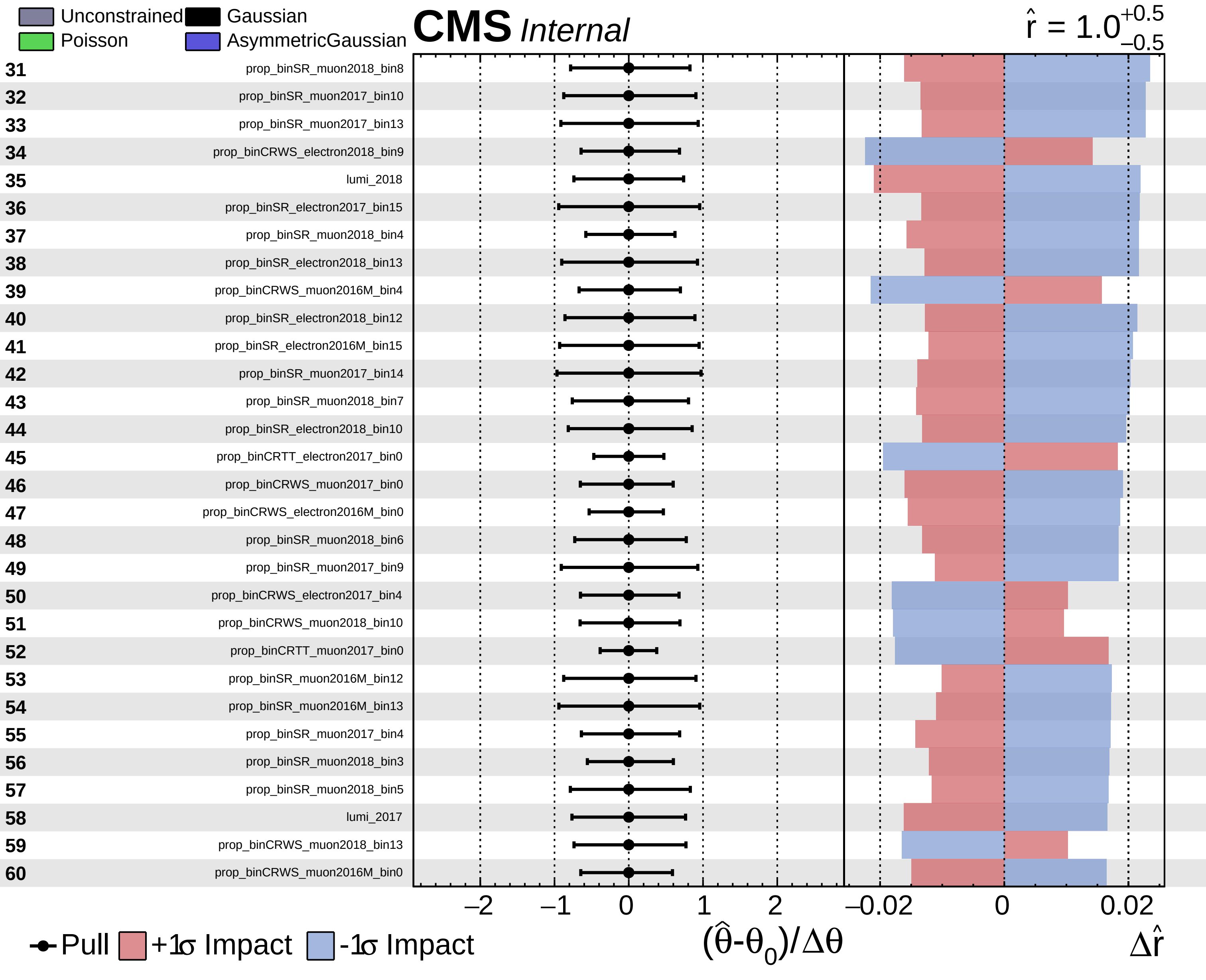
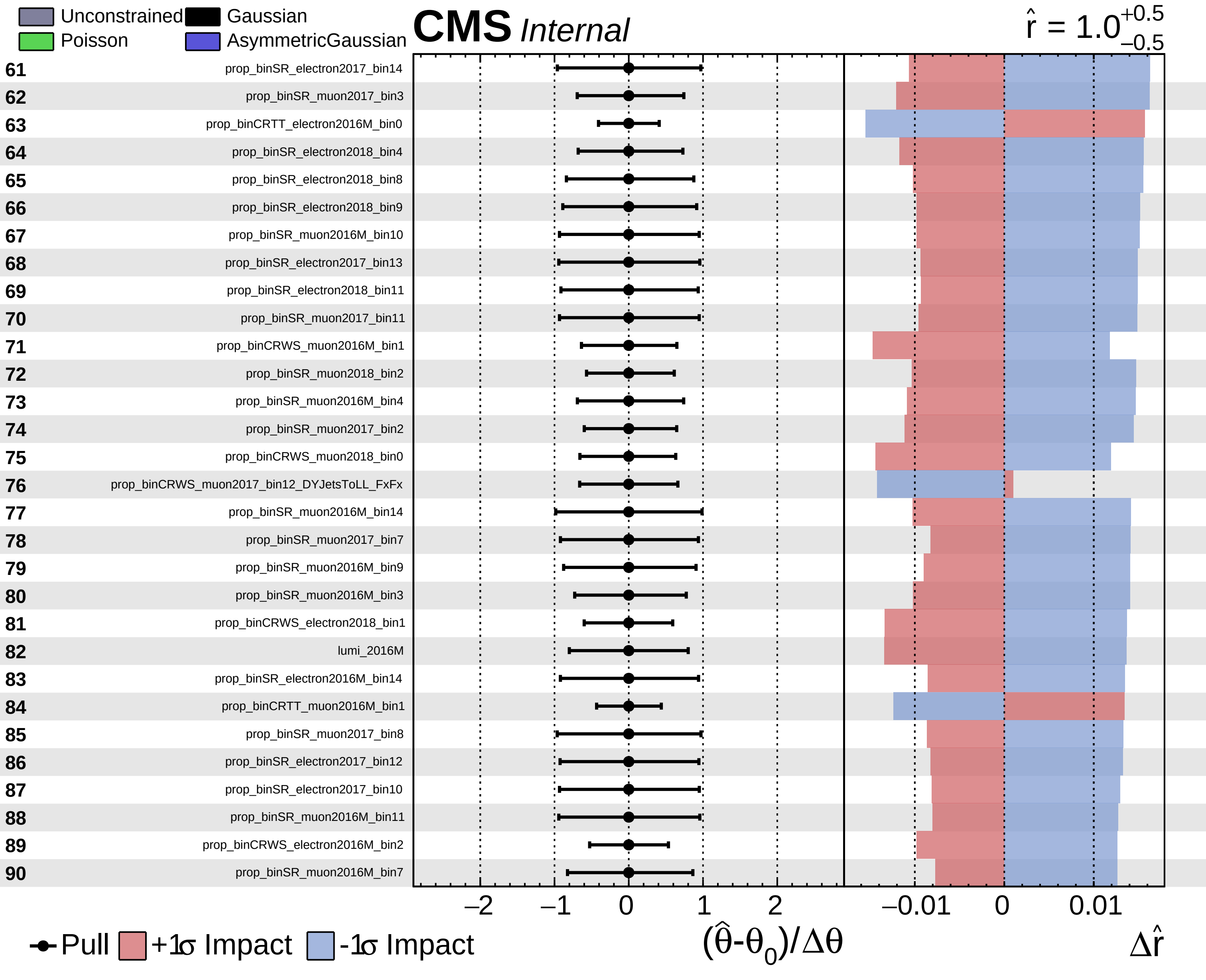
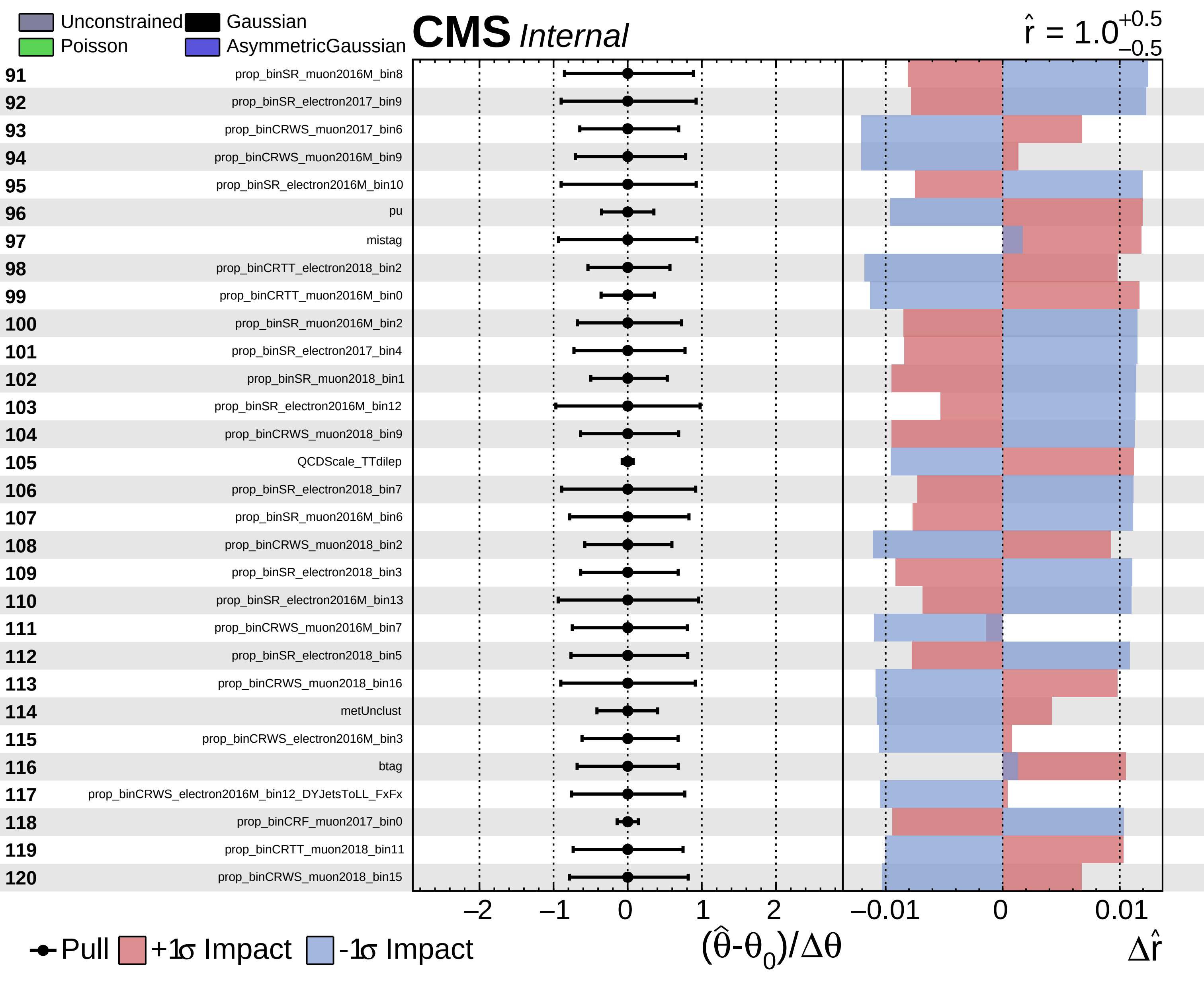


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
1 QCDScale_sig
2 prop_binSR_muon2018_bin16
3 pdf_Tot
4 prop_binSR_electron2018_bin16
5 prop_binSR_muon2018_bin15
6 prop_binSR_muon2018_bin14
7 DYest_electron_2018
$1.0^{+0.4}_{-0.4}$
8 prop_binSR_muon2017_bin16
9 DYest_muon_2017
$1.00^{+0.26}_{-0.26}$
10 prop_binSR_electron2017_bin16
11 DYest_muon_2016M
$1.00^{+0.32}_{-0.32}$
12 prop_binSR_electron2018_bin15
13 prop_binSR_muon2017_bin15
14 prop_binSR_muon2018_bin13
15 prop_binSR_muon2016M_bin16
16 DYest_muon_2018
$1.00^{+0.20}_{-0.20}$
17 prop_binSR_electron2016M_bin16
18 prop_binSR_muon2018_bin10
19 prop_binSR_muon2018_bin9
20 DYest_electron_2016M
$1.00^{+0.33}_{-0.33}$
21 prop_binCRWS_muon2016M_bin2
22 PF
23 prop_binSR_muon2018_bin12
24 DYest_electron_2017
$1.00^{+0.30}_{-0.30}$
25 prop_binCRWS_electron2018_bin5
26 prop_binSR_muon2018_bin11
27 prop_binSR_electron2018_bin14
28 prop_binCRWS_muon2018_bin3
29 prop_binSR_muon2016M_bin15
30 prop_binCRWS_muon2017_bin4
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

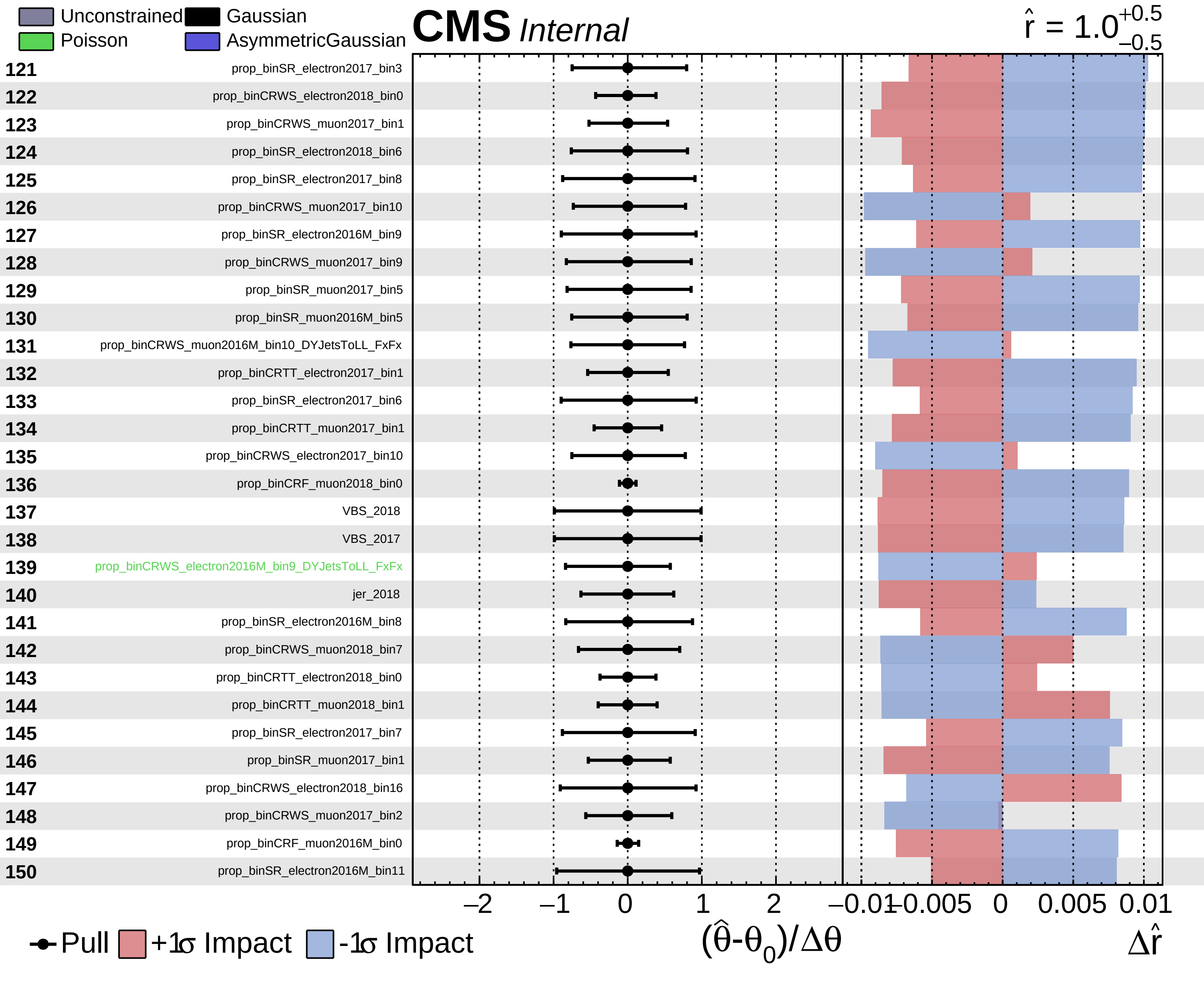


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.5
31 prop_binSR_muon2018_bin8
32 prop_binSR_muon2017_bin10
33 prop_binSR_muon2017_bin13
34 prop_binCRWS_electron2018_bin9
35 lumi_2018
36 prop_binSR_electron2017_bin15
37 prop_binSR_muon2018_bin4
38 prop_binSR_electron2018_bin13
39 prop_binCRWS_muon2016M_bin4
40 prop_binSR_electron2018_bin12
41 prop_binSR_electron2016M_bin15
42 prop_binSR_muon2017_bin14
43 prop_binSR_muon2018_bin7
44 prop_binSR_electron2018_bin10
45 prop_binCRTT_electron2017_bin0
46 prop_binCRWS_muon2017_bin0
47 prop_binCRWS_electron2016M_bin0
48 prop_binSR_muon2018_bin6
49 prop_binSR_muon2017_bin9
50 prop_binCRWS_electron2017_bin4
51 prop_binCRWS_muon2018_bin10
52 prop_binCRTT_muon2017_bin0
53 prop_binSR_muon2016M_bin12
54 prop_binSR_muon2016M_bin13
55 prop_binSR_muon2017_bin4
56 prop_binSR_muon2018_bin3
57 prop_binSR_muon2018_bin5
58 lumi_2017
59 prop_binCRWS_muon2018_bin13
60 prop_binCRWS_muon2016M_bin0
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

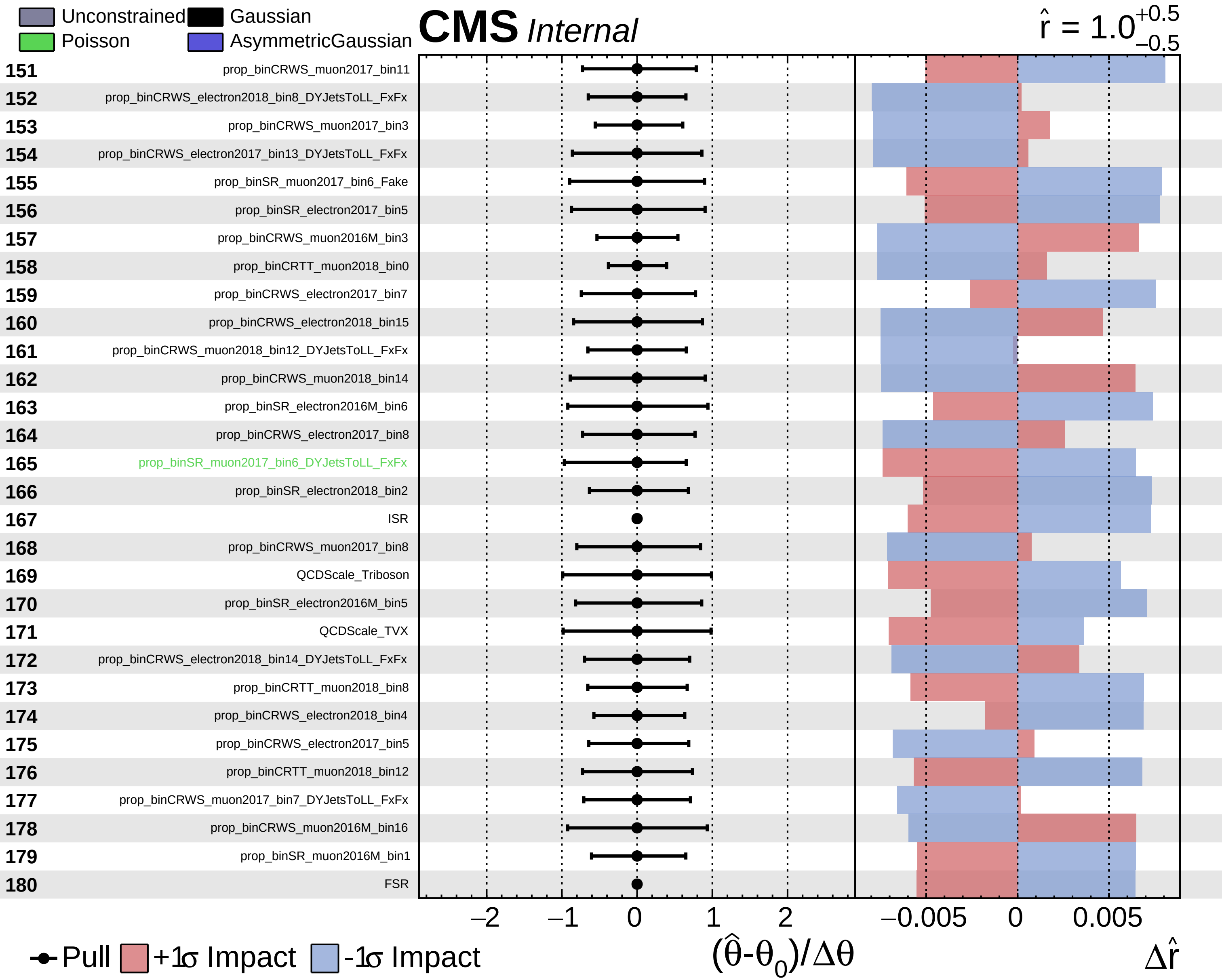

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
61 prop_binSR_electron2017_bin14
62 prop_binSR_muon2017_bin3
63 prop_binCRTT_electron2016M_bin0
64 prop_binSR_electron2018_bin4
65 prop_binSR_electron2018_bin8
66 prop_binSR_electron2018_bin9
67 prop_binSR_muon2016M_bin10
68 prop_binSR_electron2017_bin13
69 prop_binSR_electron2018_bin11
70 prop_binSR_muon2017_bin11
71 prop_binCRWS_muon2016M_bin1
72 prop_binSR_muon2018_bin2
73 prop_binSR_muon2016M_bin4
74 prop_binSR_muon2017_bin2
75 prop_binCRWS_muon2018_bin0
76 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
77 prop_binSR_muon2016M_bin14
78 prop_binSR_muon2017_bin7
79 prop_binSR_muon2016M_bin9
80 prop_binSR_muon2016M_bin3
81 prop_binCRWS_electron2018_bin1
82 lumi_2016M
83 prop_binSR_electron2016M_bin14
84 prop_binCRTT_muon2016M_bin1
85 prop_binSR_muon2017_bin8
86 prop_binSR_electron2017_bin12
87 prop_binSR_electron2017_bin10
88 prop_binSR_muon2016M_bin11
89 prop_binCRWS_electron2016M_bin2
90 prop_binSR_muon2016M_bin7
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



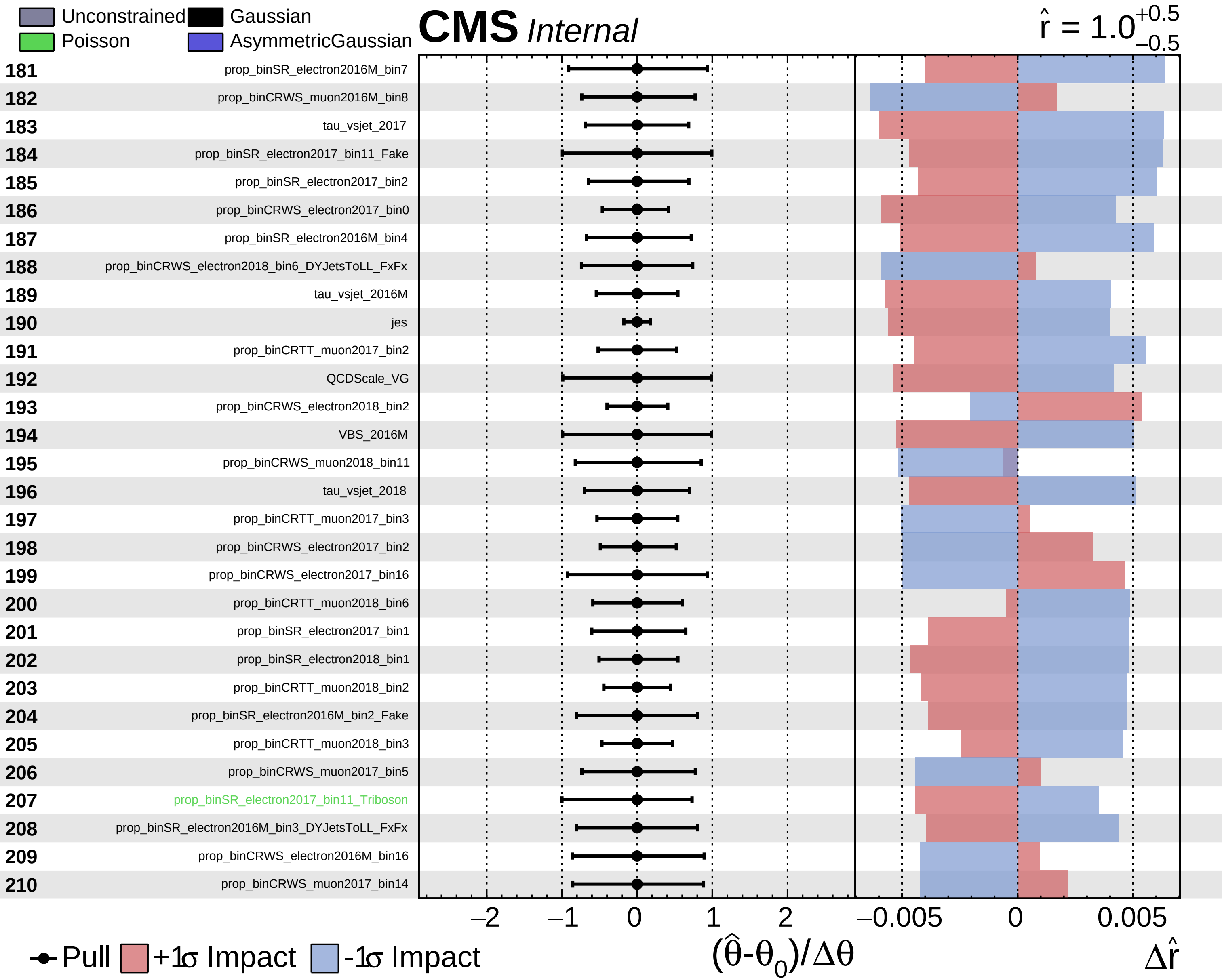
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
91 prop_binSR_muon2016M_bin8
92 prop_binSR_electron2017_bin9
93 prop_binCRWS_muon2017_bin6
94 prop_binCRWS_muon2016M_bin9
95 prop_binSR_electron2016M_bin10
96 pu
97 mistag
98 prop_binCRTT_electron2018_bin2
99 prop_binCRTT_muon2016M_bin0
100 prop_binSR_muon2016M_bin2
101 prop_binSR_electron2017_bin4
102 prop_binSR_muon2018_bin1
103 prop_binSR_electron2016M_bin12
104 prop_binCRWS_muon2018_bin9
105 QCDScale_TTdilep
106 prop_binSR_electron2018_bin7
107 prop_binSR_muon2016M_bin6
108 prop_binCRWS_muon2018_bin2
109 prop_binSR_electron2018_bin3
110 prop_binSR_electron2016M_bin13
111 prop_binCRWS_muon2016M_bin7
112 prop_binSR_electron2018_bin5
113 prop_binCRWS_muon2018_bin16
114 metUnclust
115 prop_binCRWS_electron2016M_bin3
116 btag
117 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
118 prop_binCRF_muon2017_bin0
119 prop_binCRTT_muon2018_bin11
120 prop_binCRWS_muon2018_bin15
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



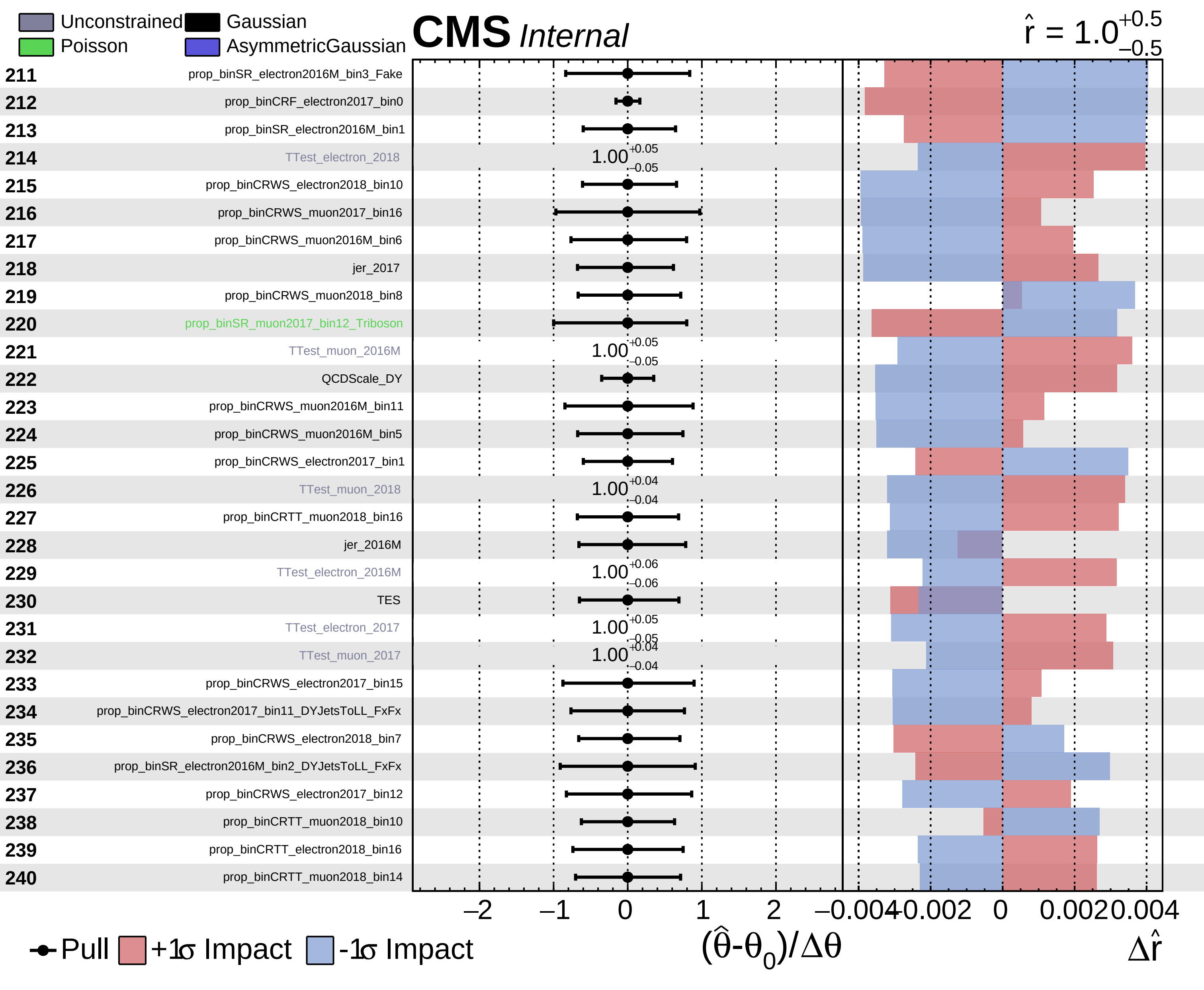
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
121 prop_binSR_electron2017_bin3
122 prop_binCRWS_electron2018_bin0
123 prop_binCRWS_muon2017_bin1
124 prop_binSR_electron2018_bin6
125 prop_binSR_electron2017_bin8
126 prop_binCRWS_muon2017_bin10
127 prop_binSR_electron2016M_bin9
128 prop_binCRWS_muon2017_bin9
129 prop_binSR_muon2017_bin5
130 prop_binSR_muon2016M_bin5
131 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
132 prop_binCRTT_electron2017_bin1
133 prop_binSR_electron2017_bin6
134 prop_binCRTT_muon2017_bin1
135 prop_binCRWS_electron2017_bin10
136 prop_binCRF_muon2018_bin0
137 VBS_2018
138 VBS_2017
139 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
140 jer_2018
141 prop_binSR_electron2016M_bin8
142 prop_binCRWS_muon2018_bin7
143 prop_binCRTT_electron2018_bin0
144 prop_binCRTT_muon2018_bin1
145 prop_binSR_electron2017_bin7
146 prop_binSR_muon2017_bin1
147 prop_binCRWS_electron2018_bin16
148 prop_binCRWS_muon2017_bin2
149 prop_binCRF_muon2016M_bin0
150 prop_binSR_electron2016M_bin11
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

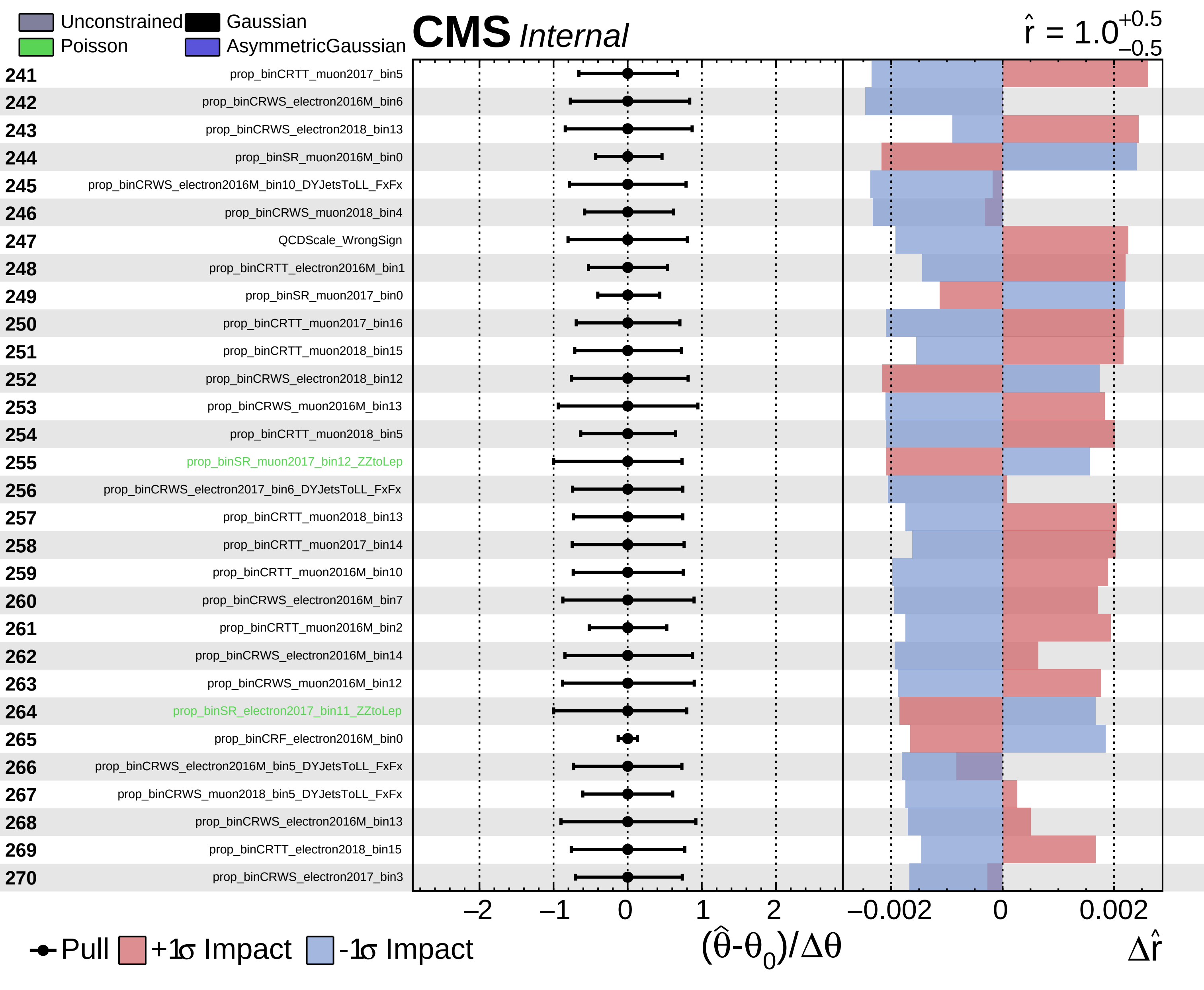


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
151 prop_binCRWS_muon2017_bin11
152 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
153 prop_binCRWS_muon2017_bin3
154 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
155 prop_binSR_muon2017_bin6_Fake
156 prop_binSR_electron2017_bin5
157 prop_binCRWS_muon2016M_bin3
158 prop_binCRTT_muon2018_bin0
159 prop_binCRWS_electron2017_bin7
160 prop_binCRWS_electron2018_bin15
161 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
162 prop_binCRWS_muon2018_bin14
163 prop_binSR_electron2016M_bin6
164 prop_binCRWS_electron2017_bin8
165 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
166 prop_binSR_electron2018_bin2
167 ISR
168 prop_binCRWS_muon2017_bin8
169 QCDScale_Triboson
170 prop_binSR_electron2016M_bin5
171 QCDScale_TVX
172 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
173 prop_binCRTT_muon2018_bin8
174 prop_binCRWS_electron2018_bin4
175 prop_binCRWS_electron2017_bin5
176 prop_binCRTT_muon2018_bin12
177 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
178 prop_binCRWS_muon2016M_bin16
179 prop_binSR_muon2016M_bin1
180 FSR
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

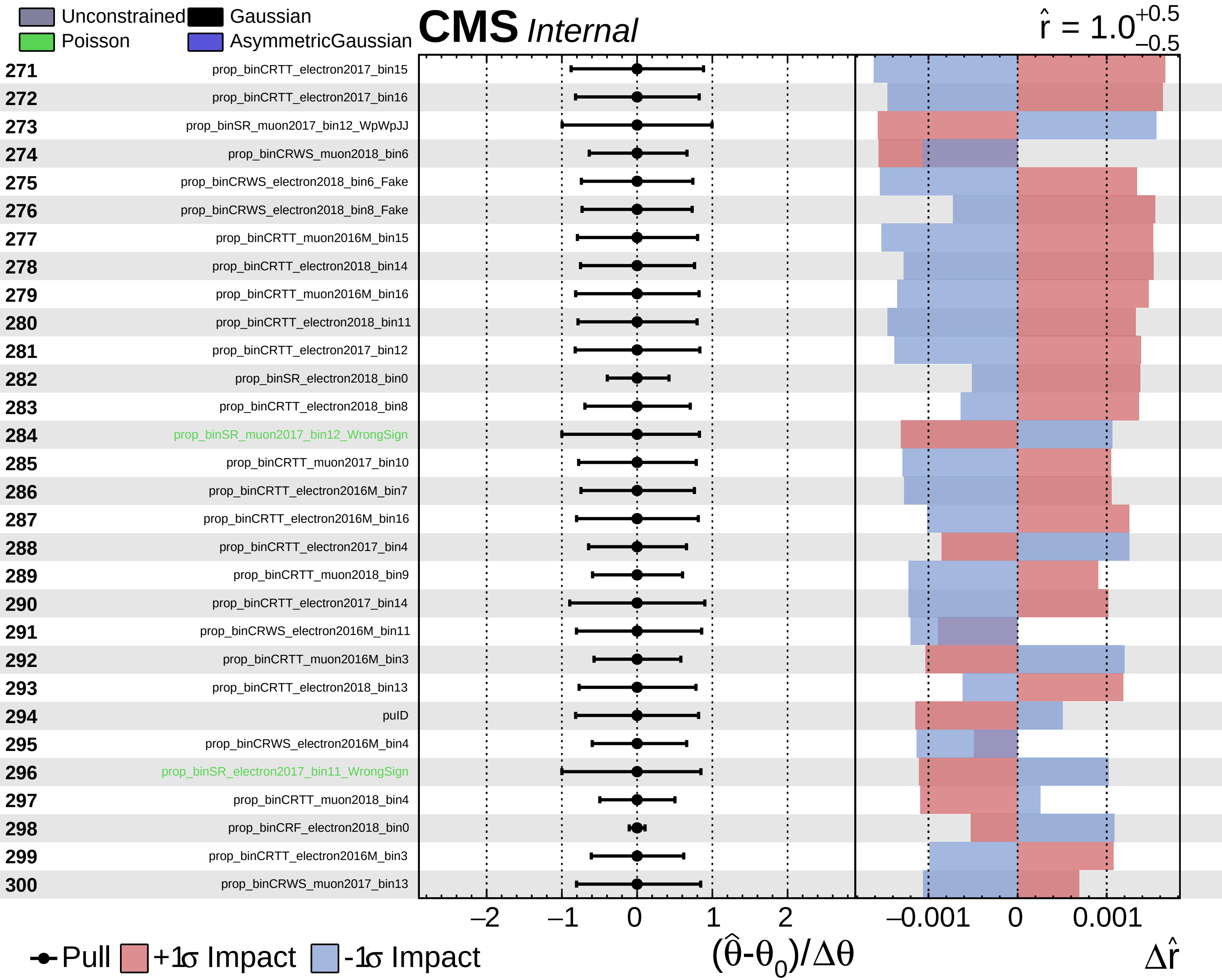


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
181 prop_binSR_electron2016M_bin7
182 prop_binCRWS_muon2016M_bin8
183 tau_vsjet_2017
184 prop_binSR_electron2017_bin11_Fake
185 prop_binSR_electron2017_bin2
186 prop_binCRWS_electron2017_bin0
187 prop_binSR_electron2016M_bin4
188 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
189 tau_vsjet_2016M
190 jes
191 prop_binCRTT_muon2017_bin2
192 QCDScale_VG
193 prop_binCRWS_electron2018_bin2
194 VBS_2016M
195 prop_binCRWS_muon2018_bin11
196 tau_vsjet_2018
197 prop_binCRTT_muon2017_bin3
198 prop_binCRWS_electron2017_bin2
199 prop_binCRWS_electron2017_bin16
200 prop_binCRTT_muon2018_bin6
201 prop_binSR_electron2017_bin1
202 prop_binSR_electron2018_bin1
203 prop_binCRTT_muon2018_bin2
204 prop_binSR_electron2016M_bin2_Fake
205 prop_binCRTT_muon2018_bin3
206 prop_binCRWS_muon2017_bin5
207 prop_binSR_electron2017_bin11_Triboson
208 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
209 prop_binCRWS_electron2016M_bin16
210 prop_binCRWS_muon2017_bin14
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

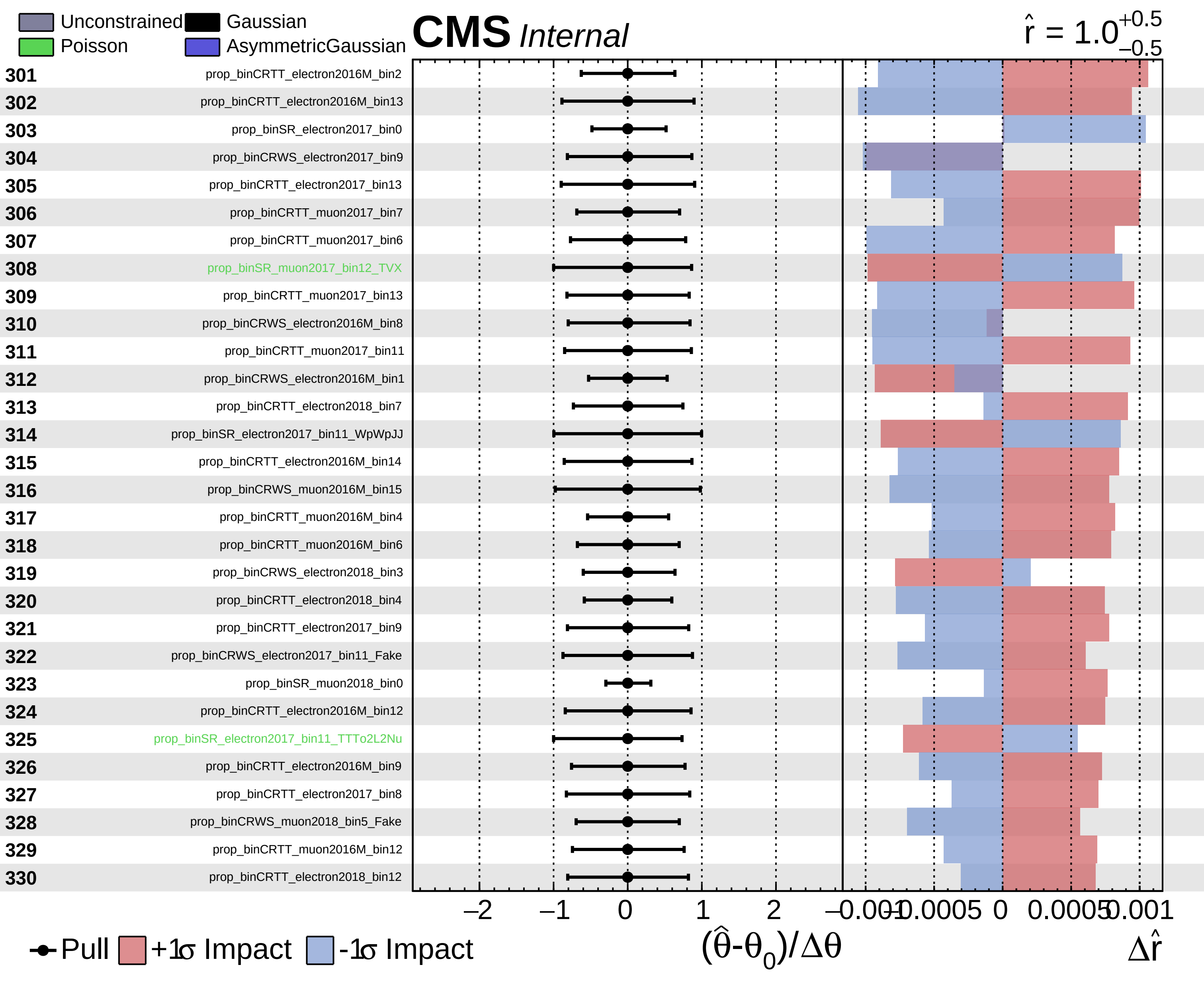

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
211 prop_binSR_electron2016M_bin3_Fake
212 prop_binCRF_electron2017_bin0
213 prop_binSR_electron2016M_bin1
214 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
215 prop_binCRWS_electron2018_bin10
216 prop_binCRWS_muon2017_bin16
217 prop_binCRWS_muon2016M_bin6
218 jer_2017
219 prop_binCRWS_muon2018_bin8
220 prop_binSR_muon2017_bin12_Triboson
221 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
222 QCDScale_DY
223 prop_binCRWS_muon2016M_bin11
224 prop_binCRWS_muon2016M_bin5
225 prop_binCRWS_electron2017_bin1
226 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
227 prop_binCRTT_muon2018_bin16
228 jer_2016M
229 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
230 TES
231 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
232 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
233 prop_binCRWS_electron2017_bin15
234 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
235 prop_binCRWS_electron2018_bin7
236 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
237 prop_binCRWS_electron2017_bin12
238 prop_binCRTT_muon2018_bin10
239 prop_binCRTT_electron2018_bin16
240 prop_binCRTT_muon2018_bin14
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

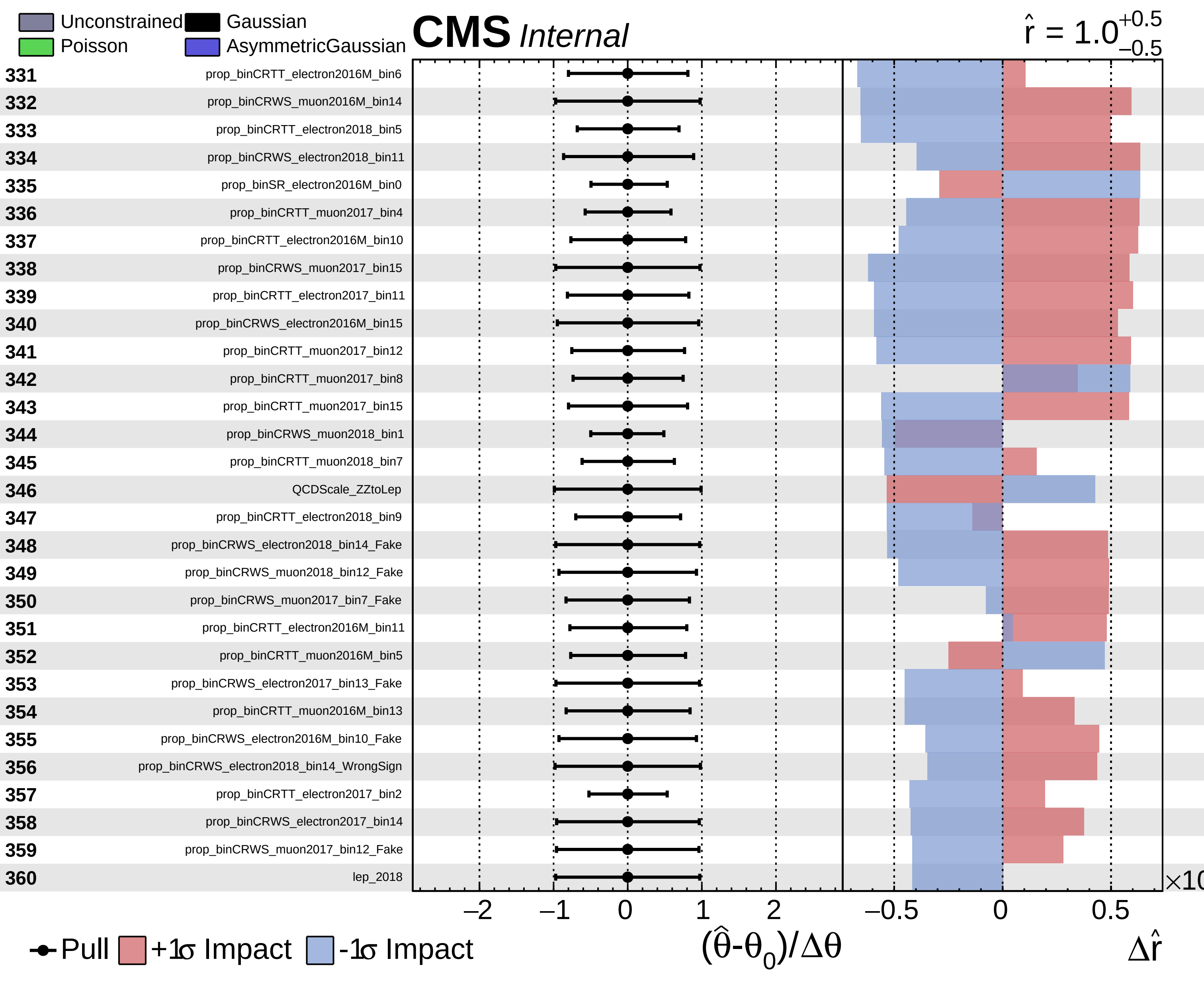


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
241 prop_binCRTT_muon2017_bin5
242 prop_binCRWS_electron2016M_bin6
243 prop_binCRWS_electron2018_bin13
244 prop_binSR_muon2016M_bin0
245 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
246 prop_binCRWS_muon2018_bin4
247 QCDScale_WrongSign
248 prop_binCRTT_electron2016M_bin1
249 prop_binSR_muon2017_bin0
250 prop_binCRTT_muon2017_bin16
251 prop_binCRTT_muon2018_bin15
252 prop_binCRWS_electron2018_bin12
253 prop_binCRWS_muon2016M_bin13
254 prop_binCRTT_muon2018_bin5
255 prop_binSR_muon2017_bin12_ZZtoLep
256 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
257 prop_binCRTT_muon2018_bin13
258 prop_binCRTT_muon2017_bin14
259 prop_binCRTT_muon2016M_bin10
260 prop_binCRWS_electron2016M_bin7
261 prop_binCRTT_muon2016M_bin2
262 prop_binCRWS_electron2016M_bin14
263 prop_binCRWS_muon2016M_bin12
264 prop_binSR_electron2017_bin11_ZZtoLep
265 prop_binCRF_electron2016M_bin0
266 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
267 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
268 prop_binCRWS_electron2016M_bin13
269 prop_binCRTT_electron2018_bin15
270 prop_binCRWS_electron2017_bin3
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

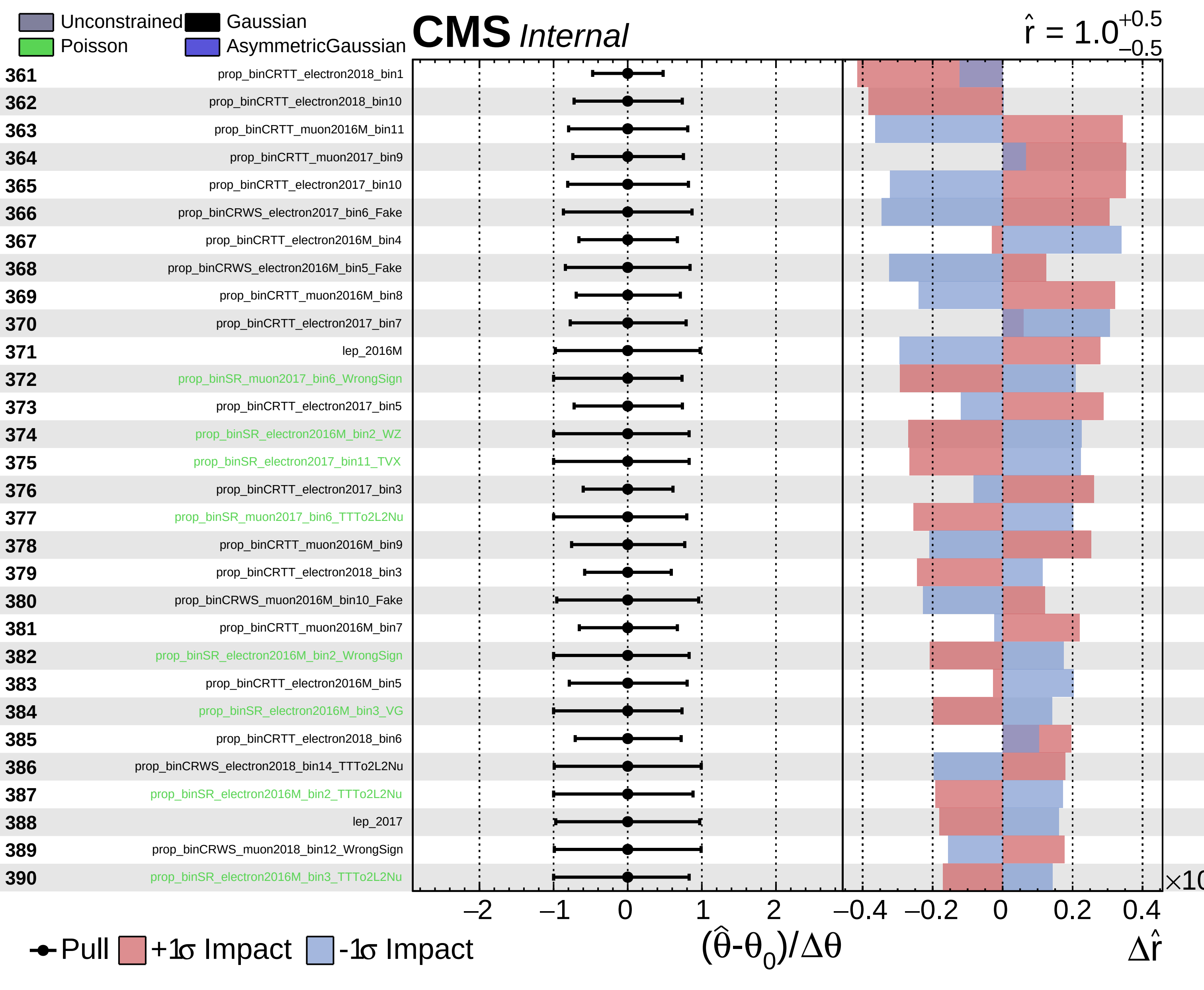


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
271 prop_binCRTT_electron2017_bin15
272 prop_binCRTT_electron2017_bin16
273 prop_binSR_muon2017_bin12_WpWpJJ
274 prop_binCRWS_muon2018_bin6
275 prop_binCRWS_electron2018_bin6_Fake
276 prop_binCRWS_electron2018_bin8_Fake
277 prop_binCRTT_muon2016M_bin15
278 prop_binCRTT_electron2018_bin14
279 prop_binCRTT_muon2016M_bin16
280 prop_binCRTT_electron2018_bin11
281 prop_binCRTT_electron2017_bin12
282 prop_binSR_electron2018_bin0
283 prop_binCRTT_electron2018_bin8
284 prop_binSR_muon2017_bin12_WrongSign
285 prop_binCRTT_muon2017_bin10
286 prop_binCRTT_electron2016M_bin7
287 prop_binCRTT_electron2016M_bin16
288 prop_binCRTT_electron2017_bin4
289 prop_binCRTT_muon2018_bin9
290 prop_binCRTT_electron2017_bin14
291 prop_binCRWS_electron2016M_bin11
292 prop_binCRTT_muon2016M_bin3
293 prop_binCRTT_electron2018_bin13
294 puID
295 prop_binCRWS_electron2016M_bin4
296 prop_binSR_electron2017_bin11_WrongSign
297 prop_binCRTT_muon2018_bin4
298 prop_binCRF_electron2018_bin0
299 prop_binCRTT_electron2016M_bin3
300 prop_binCRWS_muon2017_bin13
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

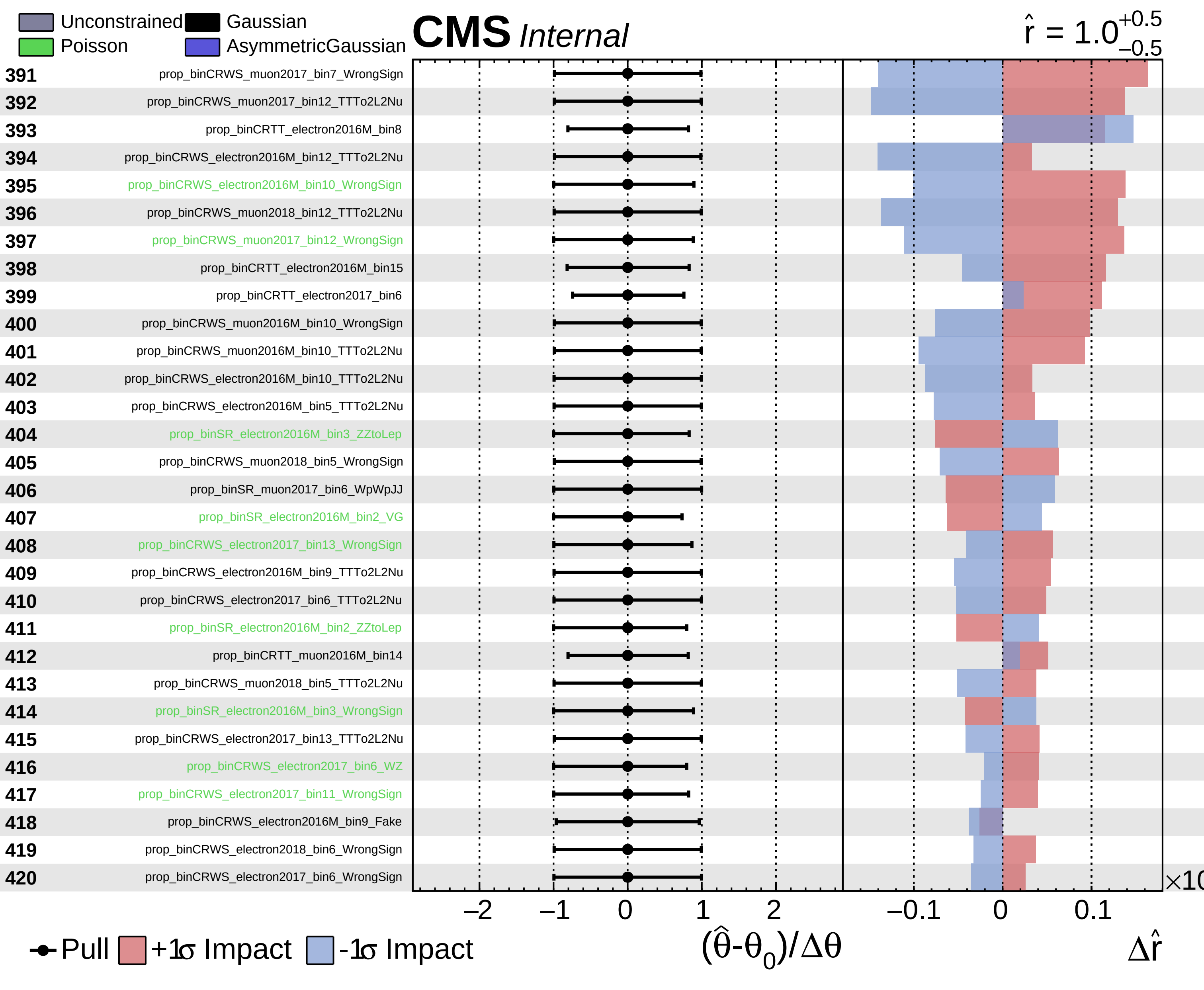

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
301 prop_binCRTT_electron2016M_bin2
302 prop_binCRTT_electron2016M_bin13
303 prop_binSR_electron2017_bin0
304 prop_binCRWS_electron2017_bin9
305 prop_binCRTT_electron2017_bin13
306 prop_binCRTT_muon2017_bin7
307 prop_binCRTT_muon2017_bin6
308 prop_binSR_muon2017_bin12_TVX
309 prop_binCRTT_muon2017_bin13
310 prop_binCRWS_electron2016M_bin8
311 prop_binCRTT_muon2017_bin11
312 prop_binCRWS_electron2016M_bin1
313 prop_binCRTT_electron2018_bin7
314 prop_binSR_electron2017_bin11_WpWpJJ
315 prop_binCRTT_electron2016M_bin14
316 prop_binCRWS_muon2016M_bin15
317 prop_binCRTT_muon2016M_bin4
318 prop_binCRTT_muon2016M_bin6
319 prop_binCRWS_electron2018_bin3
320 prop_binCRTT_electron2018_bin4
321 prop_binCRTT_electron2017_bin9
322 prop_binCRWS_electron2017_bin11_Fake
323 prop_binSR_muon2018_bin0
324 prop_binCRTT_electron2016M_bin12
325 prop_binSR_electron2017_bin11_TTTo2L2Nu
326 prop_binCRTT_electron2016M_bin9
327 prop_binCRTT_electron2017_bin8
328 prop_binCRWS_muon2018_bin5_Fake
329 prop_binCRTT_muon2016M_bin12
330 prop_binCRTT_electron2018_bin12
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



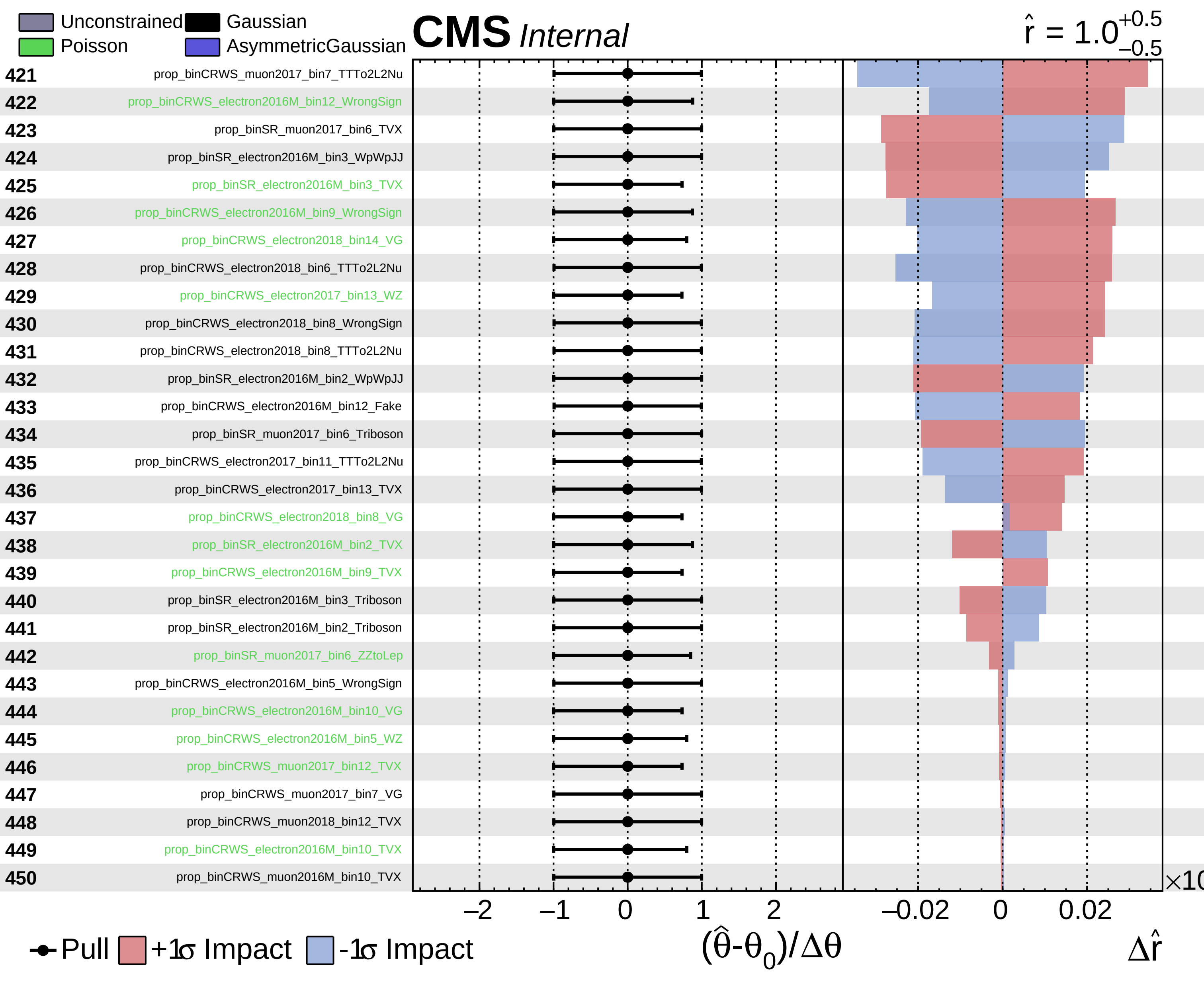
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
331 prop_binCRTT_electron2016M_bin6
332 prop_binCRWS_muon2016M_bin14
333 prop_binCRTT_electron2018_bin5
334 prop_binCRWS_electron2018_bin11
335 prop_binSR_electron2016M_bin0
336 prop_binCRTT_muon2017_bin4
337 prop_binCRTT_electron2016M_bin10
338 prop_binCRWS_muon2017_bin15
339 prop_binCRTT_electron2017_bin11
340 prop_binCRWS_electron2016M_bin15
341 prop_binCRTT_muon2017_bin12
342 prop_binCRTT_muon2017_bin8
343 prop_binCRTT_muon2017_bin15
344 prop_binCRWS_muon2018_bin1
345 prop_binCRTT_muon2018_bin7
346 QCDScale_ZZtoLep
347 prop_binCRTT_electron2018_bin9
348 prop_binCRWS_electron2018_bin14_Fake
349 prop_binCRWS_muon2018_bin12_Fake
350 prop_binCRWS_muon2017_bin7_Fake
351 prop_binCRTT_electron2016M_bin11
352 prop_binCRTT_muon2016M_bin5
353 prop_binCRWS_electron2017_bin13_Fake
354 prop_binCRTT_muon2016M_bin13
355 prop_binCRWS_electron2016M_bin10_Fake
356 prop_binCRWS_electron2018_bin14_WrongSign
357 prop_binCRTT_electron2017_bin2
358 prop_binCRWS_electron2017_bin14
359 prop_binCRWS_muon2017_bin12_Fake
360 lep_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



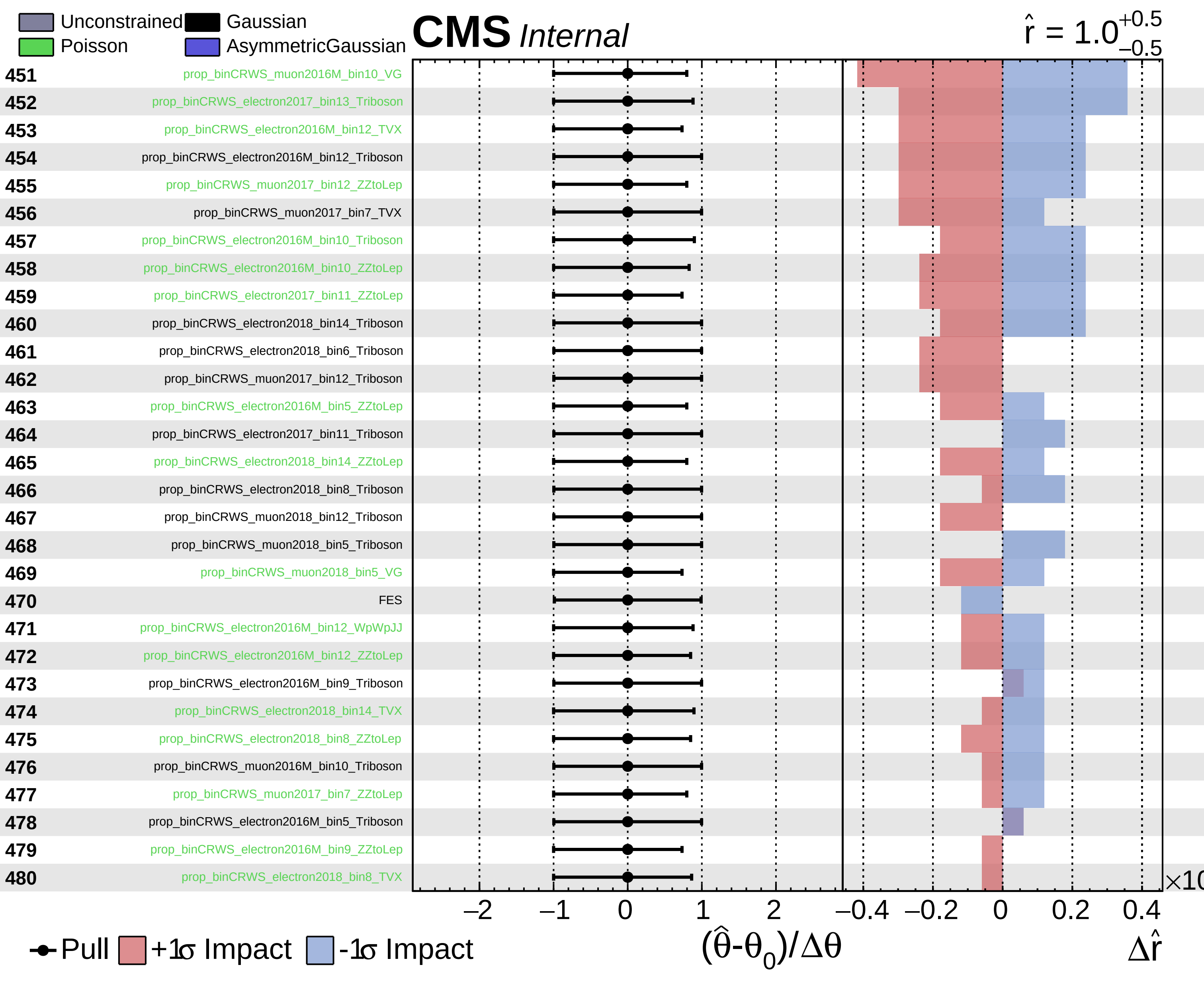
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
361 prop_binCRTT_electron2018_bin1
362 prop_binCRTT_electron2018_bin10
363 prop_binCRTT_muon2016M_bin11
364 prop_binCRTT_muon2017_bin9
365 prop_binCRTT_electron2017_bin10
366 prop_binCRWS_electron2017_bin6_Fake
367 prop_binCRTT_electron2016M_bin4
368 prop_binCRWS_electron2016M_bin5_Fake
369 prop_binCRTT_muon2016M_bin8
370 prop_binCRTT_electron2017_bin7
371 lep_2016M
372 prop_binSR_muon2017_bin6_WrongSign
373 prop_binCRTT_electron2017_bin5
374 prop_binSR_electron2016M_bin2_WZ
375 prop_binSR_electron2017_bin11_TVX
376 prop_binCRTT_electron2017_bin3
377 prop_binSR_muon2017_bin6_TTTo2L2Nu
378 prop_binCRTT_muon2016M_bin9
379 prop_binCRTT_electron2018_bin3
380 prop_binCRWS_muon2016M_bin10_Fake
381 prop_binCRTT_muon2016M_bin7
382 prop_binSR_electron2016M_bin2_WrongSign
383 prop_binCRTT_electron2016M_bin5
384 prop_binSR_electron2016M_bin3_VG
385 prop_binCRTT_electron2018_bin6
386 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
387 prop_binSR_electron2016M_bin2_TTTo2L2Nu
388 lep_2017
389 prop_binCRWS_muon2018_bin12_WrongSign
390 prop_binSR_electron2016M_bin3_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



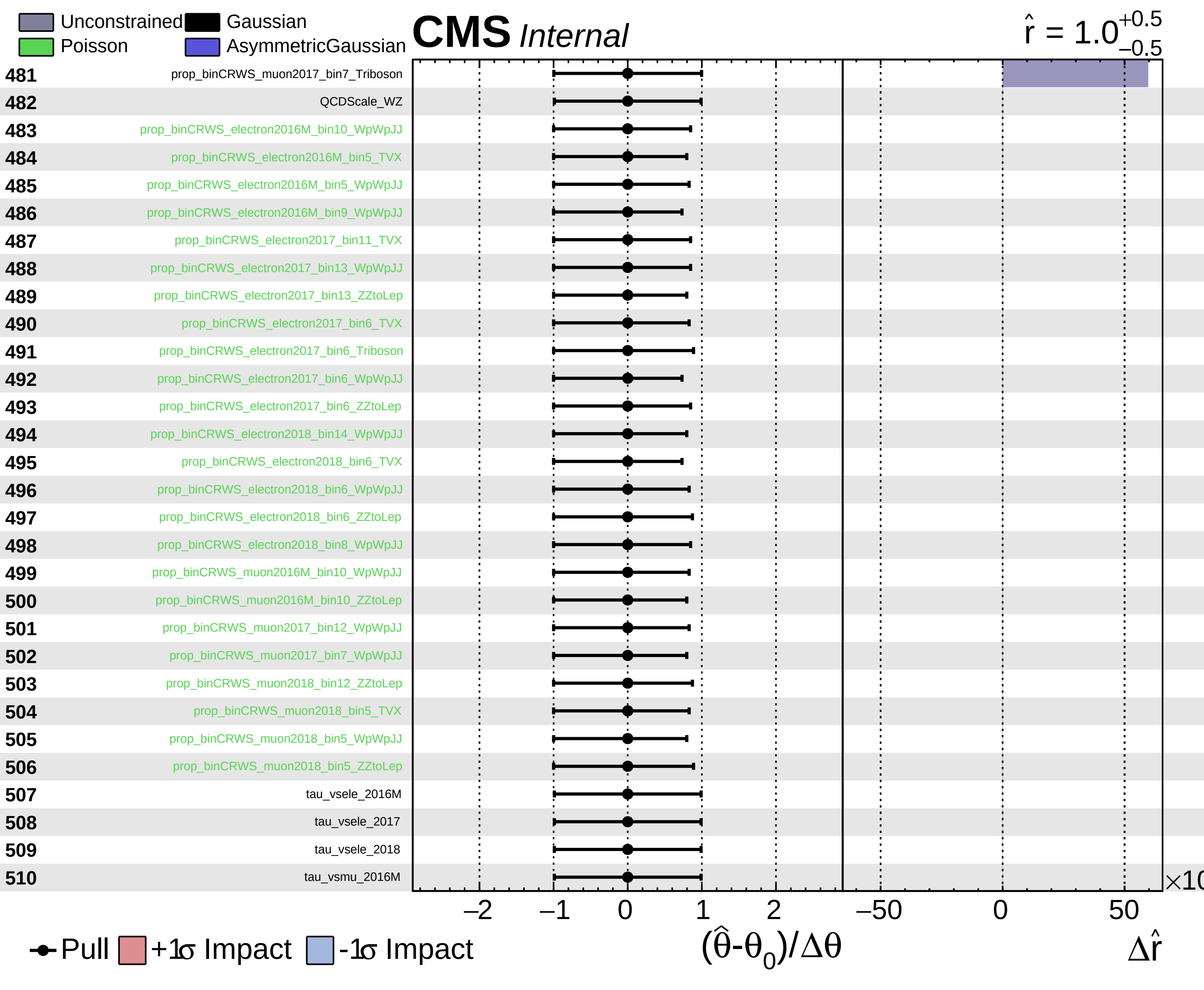
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
391 prop_binCRWS_muon2017_bin7_WrongSign
392 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
393 prop_binCRTT_electron2016M_bin8
394 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
395 prop_binCRWS_electron2016M_bin10_WrongSign
396 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
397 prop_binCRWS_muon2017_bin12_WrongSign
398 prop_binCRTT_electron2016M_bin15
399 prop_binCRTT_electron2017_bin6
400 prop_binCRWS_muon2016M_bin10_WrongSign
401 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
402 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
403 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
404 prop_binSR_electron2016M_bin3_ZZtoLep
405 prop_binCRWS_muon2018_bin5_WrongSign
406 prop_binSR_muon2017_bin6_WpWpJJ
407 prop_binSR_electron2016M_bin2_VG
408 prop_binCRWS_electron2017_bin13_WrongSign
409 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
410 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
411 prop_binSR_electron2016M_bin2_ZZtoLep
412 prop_binCRTT_muon2016M_bin14
413 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
414 prop_binSR_electron2016M_bin3_WrongSign
415 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
416 prop_binCRWS_electron2017_bin6_WZ
417 prop_binCRWS_electron2017_bin11_WrongSign
418 prop_binCRWS_electron2016M_bin9_Fake
419 prop_binCRWS_electron2018_bin6_WrongSign
420 prop_binCRWS_electron2017_bin6_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



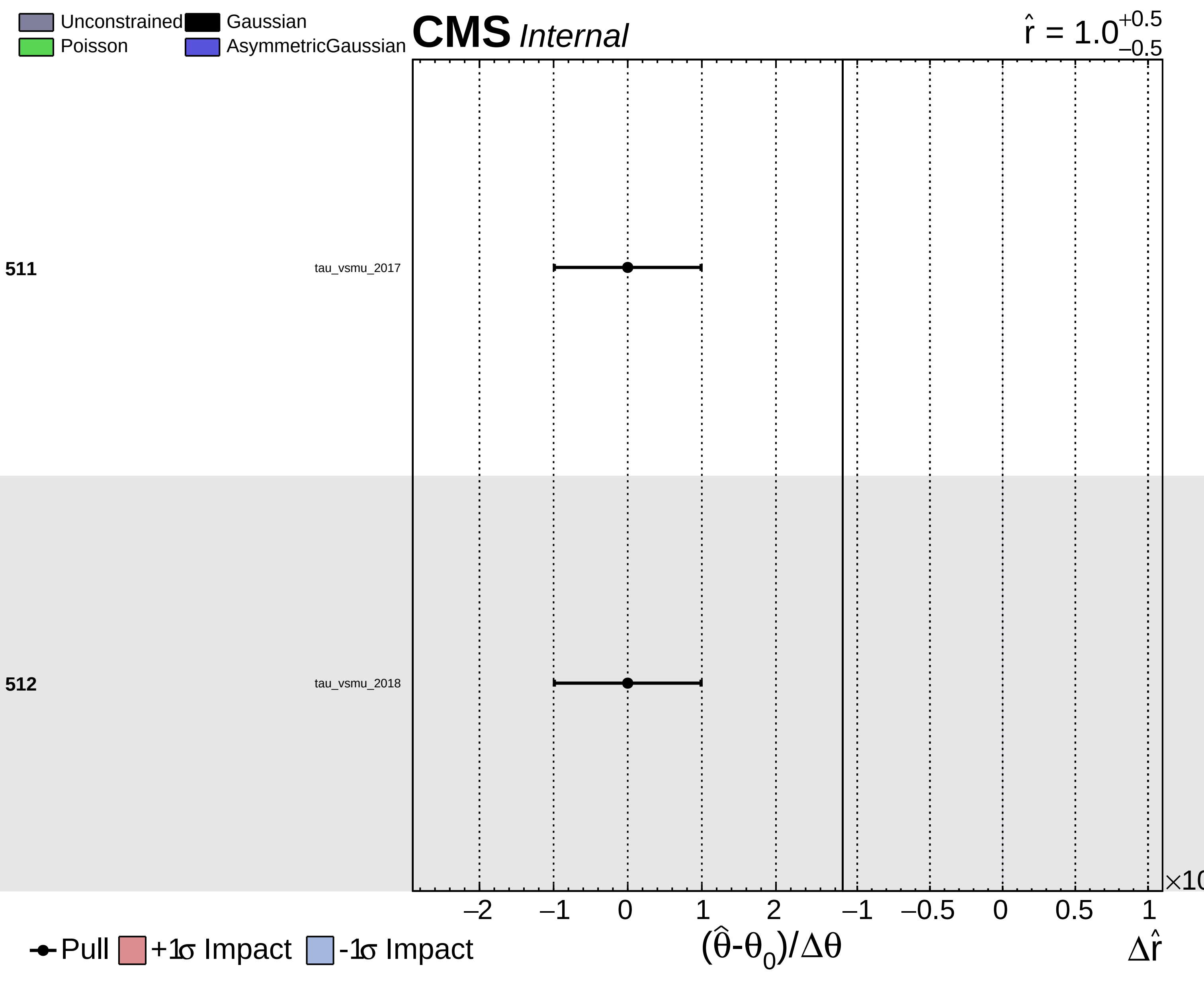
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
421 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
422 prop_binCRWS_electron2016M_bin12_WrongSign
423 prop_binSR_muon2017_bin6_TVX
424 prop_binSR_electron2016M_bin3_WpWpJJ
425 prop_binSR_electron2016M_bin3_TVX
426 prop_binCRWS_electron2016M_bin9_WrongSign
427 prop_binCRWS_electron2018_bin14_VG
428 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
429 prop_binCRWS_electron2017_bin13_WZ
430 prop_binCRWS_electron2018_bin8_WrongSign
431 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
432 prop_binSR_electron2016M_bin2_WpWpJJ
433 prop_binCRWS_electron2016M_bin12_Fake
434 prop_binSR_muon2017_bin6_Triboson
435 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
436 prop_binCRWS_electron2017_bin13_TVX
437 prop_binCRWS_electron2018_bin8_VG
438 prop_binSR_electron2016M_bin2_TVX
439 prop_binCRWS_electron2016M_bin9_TVX
440 prop_binSR_electron2016M_bin3_Triboson
441 prop_binSR_electron2016M_bin2_Triboson
442 prop_binSR_muon2017_bin6_ZZtoLep
443 prop_binCRWS_electron2016M_bin5_WrongSign
444 prop_binCRWS_electron2016M_bin10_VG
445 prop_binCRWS_electron2016M_bin5_WZ
446 prop_binCRWS_muon2017_bin12_TVX
447 prop_binCRWS_muon2017_bin7_VG
448 prop_binCRWS_muon2018_bin12_TVX
449 prop_binCRWS_electron2016M_bin10_TVX
450 prop_binCRWS_muon2016M_bin10_TVX
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
451 prop_binCRWS_muon2016M_bin10_VG
452 prop_binCRWS_electron2017_bin13_Triboson
453 prop_binCRWS_electron2016M_bin12_TVX
454 prop_binCRWS_electron2016M_bin12_Triboson
455 prop_binCRWS_muon2017_bin12_ZZtoLep
456 prop_binCRWS_muon2017_bin7_TVX
457 prop_binCRWS_electron2016M_bin10_Triboson
458 prop_binCRWS_electron2016M_bin10_ZZtoLep
459 prop_binCRWS_electron2017_bin11_ZZtoLep
460 prop_binCRWS_electron2018_bin14_Triboson
461 prop_binCRWS_electron2018_bin6_Triboson
462 prop_binCRWS_muon2017_bin12_Triboson
463 prop_binCRWS_electron2016M_bin5_ZZtoLep
464 prop_binCRWS_electron2017_bin11_Triboson
465 prop_binCRWS_electron2018_bin14_ZZtoLep
466 prop_binCRWS_electron2018_bin8_Triboson
467 prop_binCRWS_muon2018_bin12_Triboson
468 prop_binCRWS_muon2018_bin5_Triboson
469 prop_binCRWS_muon2018_bin5_VG
470 FES
471 prop_binCRWS_electron2016M_bin12_WpWpJJ
472 prop_binCRWS_electron2016M_bin12_ZZtoLep
473 prop_binCRWS_electron2016M_bin9_Triboson
474 prop_binCRWS_electron2018_bin14_TVX
475 prop_binCRWS_electron2018_bin8_ZZtoLep
476 prop_binCRWS_muon2016M_bin10_Triboson
477 prop_binCRWS_muon2017_bin7_ZZtoLep
478 prop_binCRWS_electron2016M_bin5_Triboson
479 prop_binCRWS_electron2016M_bin9_ZZtoLep
480 prop_binCRWS_electron2018_bin8_TVX
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
481 prop_binCRWS_muon2017_bin7_Triboson
482 QCDScale_WZ
483 prop_binCRWS_electron2016M_bin10_WpWpJJ
484 prop_binCRWS_electron2016M_bin5_TVX
485 prop_binCRWS_electron2016M_bin5_WpWpJJ
486 prop_binCRWS_electron2016M_bin9_WpWpJJ
487 prop_binCRWS_electron2017_bin11_TVX
488 prop_binCRWS_electron2017_bin13_WpWpJJ
489 prop_binCRWS_electron2017_bin13_ZZtoLep
490 prop_binCRWS_electron2017_bin6_TVX
491 prop_binCRWS_electron2017_bin6_Triboson
492 prop_binCRWS_electron2017_bin6_WpWpJJ
493 prop_binCRWS_electron2017_bin6_ZZtoLep
494 prop_binCRWS_electron2018_bin14_WpWpJJ
495 prop_binCRWS_electron2018_bin6_TVX
496 prop_binCRWS_electron2018_bin6_WpWpJJ
497 prop_binCRWS_electron2018_bin6_ZZtoLep
498 prop_binCRWS_electron2018_bin8_WpWpJJ
499 prop_binCRWS_muon2016M_bin10_WpWpJJ
500 prop_binCRWS_muon2016M_bin10_ZZtoLep
501 prop_binCRWS_muon2017_bin12_WpWpJJ
502 prop_binCRWS_muon2017_bin7_WpWpJJ
503 prop_binCRWS_muon2018_bin12_ZZtoLep
504 prop_binCRWS_muon2018_bin5_TVX
505 prop_binCRWS_muon2018_bin5_WpWpJJ
506 prop_binCRWS_muon2018_bin5_ZZtoLep
507 tau_vsele_2016M
508 tau_vsele_2017
509 tau_vsele_2018
510 tau_vsmu_2016M
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
511
tau_vsmu_2017
512
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$